

Week 6 — Law of Large Numbers, Duality, Conic Programming

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Oct 3, 2026

1. Law of large numbers

$\tilde{\xi}_1, \tilde{\xi}_2, \tilde{\xi}_3, \dots$: an infinite sequence of **independent and identically distributed (i.i.d.)** random variables,

each with expected value μ

Law of large numbers (strong law, SLLN)

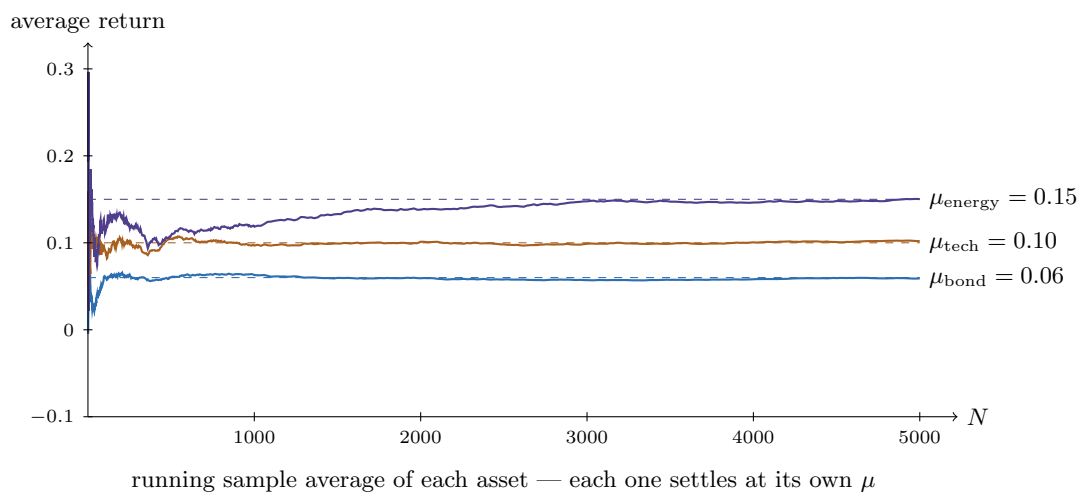
$$\mathbb{P}\left(\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i \in [N]} \tilde{\xi}_i = \mu\right) = 1$$

sample average $\longrightarrow$ expected value

Example portfolio: $\tilde{\xi}_i$ = return of an asset on day i , i.i.d. over days

asset	expected return μ	σ
bond	0.06	0.10
tech	0.10	0.20
energy	0.15	0.30

$$\frac{1}{N} \sum_{i \in [N]} \tilde{\xi}_i = \text{average return over the first } N \text{ days}$$



2. Duality theory

in this course, we focus on **convex** functions.

Primal Problem

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & f_i(x) \leq 0 \quad \forall i \in [I] \end{aligned} \tag{P}$$

Dual Problem

$$\begin{aligned} \max_{\theta} \min_{x \in \mathbb{R}^N} \quad & f_0(x) + \sum_{i \in [I]} \theta_i f_i(x) \\ \text{s.t.} \quad & \theta \in \mathbb{R}^I \\ & \theta \geq 0 \end{aligned} \tag{D}$$

$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & f_i(x) \leq 0 \quad \forall i \in [I] \end{aligned}$
$= \min_{x \in \mathbb{R}^N} f_0(x) + \underbrace{\max_{\theta \in \mathbb{R}_+^I} \sum_{i \in [I]} \theta_i f_i(x)}_{\leq 0}$

if $f_i(x) \leq 0 \quad \forall i$, then $= 0$

if $\exists i : f_i(x) > 0$, then $= +\infty$

$$\geq \max_{\theta \in \mathbb{R}_+^I} \min_{x \in \mathbb{R}^N} f_0(x) + \sum_{i \in [I]} \theta_i f_i(x)$$

Properties:

— **weak duality** $\min(P) \geq \max(D)$ always holds.

— **strong duality:** $\min(P) = \max(D)$ holds

when the problem (P) is convex and the Slater's condition holds

$$\exists x \in \mathbb{R}^N, \text{ s.t. } f_i(x) < 0 \quad \forall i \in [I]$$

In this course, we assume that strong duality holds for convex problem!!

3. Example: linear program

Primal LP:

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & x \in \mathbb{R}^N \\ & Ax \leq b \end{array} \quad \begin{array}{ll} f_0(x) = c^\top x \\ f_i(x) = a_i^\top x - b_i & \forall i \in [I] \end{array}$$

$$\begin{aligned} f_i(x) &\leq 0 \\ \iff a_i^\top x - b_i &\leq 0 \\ \iff a_i^\top x &\leq b_i \end{aligned}$$

Dual LP:

$$\begin{aligned} & \max_{\theta \in \mathbb{R}_+^I} \min_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i (a_i^\top x - b_i) \\ &= \max_{\theta \in \mathbb{R}_+^I} \min_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i a_i^\top x - \sum_{i \in [I]} \theta_i b_i \\ &= \max_{\theta \in \mathbb{R}_+^I} - \sum_{i \in [I]} \theta_i b_i + \underbrace{\min_{x \in \mathbb{R}^N} \left(c + \sum_{i \in [I]} \theta_i a_i \right)^\top x}_{= \begin{cases} 0 & \text{if } c + \sum_{i \in [I]} \theta_i a_i = 0 \\ -\infty & \text{if otherwise.} \end{cases}} \\ &= \max_{\substack{\theta \in \mathbb{R}_+^I \\ A^\top \theta = -c}} -b^\top \theta \end{aligned}$$

LP strong duality : if there exists $x \in \mathbb{R}^N : Ax \leq b$

$$\text{or } \theta \in \mathbb{R}^I : A^\top \theta = -c, \theta \geq 0$$

then strong duality holds

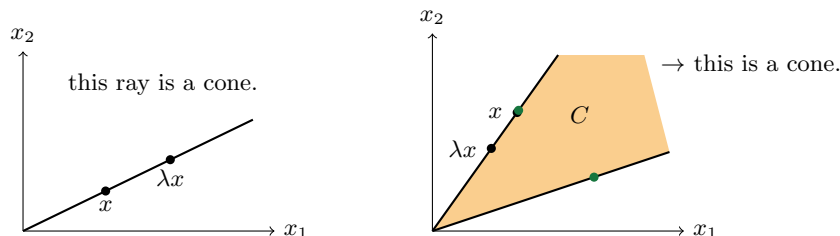
4. Conic programming

a conic program is a convex optimization problem.

— a **linear** objective function.

— a feasible set defines through the intersection of halfspaces $Ax \leq b$ or a proper cone C .

Def (cone) a set C is a cone if $\forall x \in C$, we have $\lambda x \in C \quad \forall \lambda \geq 0$



Def (proper cone) $\mathbb{R}^N$ is a cone but not pointed.

C is a proper cone if it's **convex, closed, pointed**

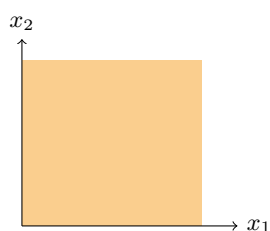
if $x, -x \in C$ then $x = 0$.

Generic conic program:

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \preceq_C b \end{aligned} \quad \Longleftrightarrow \quad b - Ax \in C$$

Example:

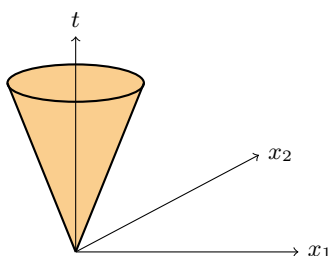
1) **Linear program:** $C = \mathbb{R}_+^K$



$$\begin{aligned} b - Ax &\in \mathbb{R}_+^K \\ \Longleftrightarrow b - Ax &\geq 0 \quad \Longleftrightarrow Ax \leq b \end{aligned}$$

2) **Second-order cone:** $C = \text{SOC}_Q^{N+1} = \{(x, t) \in \mathbb{R}^N \times \mathbb{R} : \sqrt{x^\top Q x} \leq t\}, \quad Q \succ 0$

$Q = I$: $\text{SOC}^{N+1} = \{(x, t) : \|x\|_2 \leq t\}$ “ice-cream” cone



3) **Semidefinite program:** $C = \mathbb{S}_+^K$

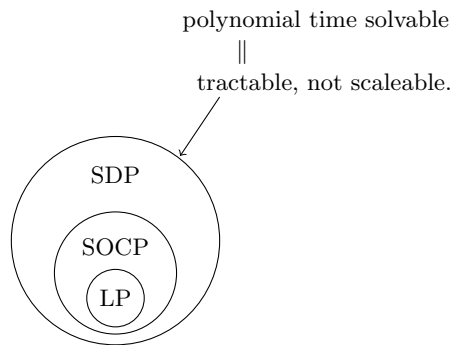
$\mathbb{S}^K$: set of symmetric matrices in $\mathbb{R}^{K \times K}$ ($A^\top = A$)

$\mathbb{S}_+^K$: set of positive semidefinite (symmetric) matrices in $\mathbb{R}^{K \times K}$

$$A \in \mathbb{S}_+^K \text{ if } z^\top A z \geq 0 \quad \forall z \in \mathbb{R}^K$$

Generic semidefinite program:

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & A_1 x_1 + A_2 x_2 + \cdots + A_N x_N \preceq_{\mathbb{S}_+^K} B \end{aligned} \quad \rightarrow \text{usually ignored}$$



4) **Several cones at once:** $C = C_1 \times C_2$ is again a proper cone

$$b_1 - A_1 x \in C_1 \text{ and } b_2 - A_2 x \in C_2 \quad \Longleftrightarrow \quad \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} - \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} x \in C_1 \times C_2$$

Example: portfolio, minimum variance

$$\begin{aligned} \min \quad & x^\top \Sigma x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & \mu^\top x \geq r \\ & e^\top x \leq 1 \\ & x \geq 0 \end{aligned} \quad \Longleftrightarrow \quad \begin{aligned} \min \quad & t \\ \text{s.t.} \quad & (x, t) \in \mathbb{R}^N \times \mathbb{R} \\ & \sqrt{x^\top \Sigma x} \leq t \\ & \mu^\top x \geq r \\ & e^\top x \leq 1 \\ & x \geq 0 \end{aligned} \quad \begin{aligned} & \Longleftrightarrow (x, t) \in \text{SOC}_{\Sigma}^{N+1} \\ & \Longleftrightarrow \mu^\top x - r \in \mathbb{R}_+ \\ & \Longleftrightarrow 1 - e^\top x \in \mathbb{R}_+ \\ & \Longleftrightarrow x \in \mathbb{R}_+^N \end{aligned}$$

$\sqrt{\cdot}$ is increasing: $\min x^\top \Sigma x$ and $\min \sqrt{x^\top \Sigma x}$ have the same minimizer x^*

t is a new variable: at the optimum $t^* = \sqrt{x^{*\top} \Sigma x^*}$, the standard deviation of the return

$\Rightarrow C = \mathbb{R}_+^{N+2} \times \text{SOC}_{\Sigma}^{N+1}$ — a SOCP in (x, t)

the coding session, $N = 2$: $\mu = \begin{bmatrix} 0.10 \\ 0.15 \end{bmatrix}$, $\Sigma = \begin{bmatrix} 0.01 & 0.001 \\ 0.001 & 0.04 \end{bmatrix}$, $r = 0.13$

$$\underbrace{\begin{bmatrix} 0.10x_1 + 0.15x_2 - 0.13 \\ 1 - x_1 - x_2 \\ x_1 \\ x_2 \end{bmatrix}}_{\in \mathbb{R}_+^4} \quad \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ t \end{bmatrix}}_{\in \text{SOC}_{\Sigma}^3} \quad \Rightarrow \quad x^* = (0.4, 0.6), \quad t^* = 0.128$$

5. Conic programming duality

Def (*dual cone*)

$$C^* = \{ y \in \mathbb{R}^K : \langle y, x \rangle \geq 0 \quad \forall x \in C \}$$

Self dual cones ($C = C^*$)

— $C = \mathbb{R}_+^K$

— $C = \text{SOC}^K$ (second-order cone, $Q = I$) general $Q \succ 0$: $(\text{SOC}_Q^K)^* = \text{SOC}_{Q^{-1}}^K$ — not self-dual

— $C = \mathbb{S}_+^K$

— products of these: $C = C_1 \times C_2, \quad C^* = C_1^* \times C_2^* = C$

Primal conic program:

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & x \in \mathbb{R}^N \\ & Ax \preceq_C b \end{array}$$

Dual conic program:

$$\begin{array}{ll} \max & -\theta^\top b \\ \text{s.t.} & \theta \in C^* \\ & A^\top \theta = -c \end{array}$$